

TPC GROUP 14

Data Modelling

Firebase Realtime Database design

Client: SkillsTrack Training Centre

Project: Learner Support Portal

Programme: Technical Programming Cycle | September 2026

Data Modelling

Purpose

This document describes the Firebase Realtime Database data model for the Learner Support Portal. Firebase stores data as a JSON tree, not SQL tables. Each top-level node is a collection. Record keys are unique IDs.

Database tree

```
Students Portal (Realtime Database)
|-- users/           who can log in (learner / assessor)
|-- tasks/           learning tasks
|-- bookings/        support session requests
|-- resources/        guides, documents and links
|-- gameScores/       coding mini-game results
|-- preferences/      theme and filter settings
```

Relationships

```
users (1) ---- (many) tasks           via userId
users (1) ---- (many) gameScores       via userId
users (1) ---- (1)   preferences       uid is the record key
users (learner) -- (many) bookings     via learnerId
users (assessor) -- (many) bookings    via assessorId
resources -- shared (no owner field; assessors write, signed-in users read)
```

Progress is not stored as a field. It is calculated from tasks: completed tasks divided by total tasks.

Collection: users

One profile per person. The record key should be the Firebase Authentication uid.

Required fields: email, displayName, role, createdAt.

Role values: learner or assessor.

Collection: tasks

Learning work items owned by a learner.

Required fields: userId, title, category, dueDate, priority, status, createdAt, updatedAt.

Priority values: low, medium, high.

Status values: todo, in-progress, done.

Collection: bookings

Support session requests linking a learner to an assessor.

Required fields: learnerId, topic, preferredDate, status, createdAt.

Optional fields: assessorId, notes.

Status values: pending, confirmed, completed, cancelled.

Collection: resources

Learning materials uploaded by an assessor.

Required fields: title, type, url, description, category.

Type values: document, link, guide.

Collection: gameScores

Mini-game results for a learner.

Required fields: userId, score, playedAt.

Optional field: level.

Collection: preferences

UI preferences keyed by uid.

Required field: theme (light or dark).

Optional field: taskFilter.

Security (role-based)

- A learner may read and write their own profile, tasks, bookings, scores and preferences.
- An assessor may read learner profiles, tasks, bookings and scores, update booking status, and upload resources.
- Passwords are never stored in this database. Firebase Authentication holds login credentials.

Sample records

Demo learner: learnerDemo001 (Lucia Ashler Mahlangu, role learner).

Demo assessor: assessorDemo001 (Londeka Zikalala, role assessor).

Tasks task001 and task002 belong to learnerDemo001.

Booking booking001 links learnerDemo001 to assessorDemo001.

Field summary

Users

Field	Type / values	Notes
email	string	Login email (not the password)
displayName	string	Full name shown in the portal
role	learner assessor	Controls dashboard and security rules
createdAt	ISO date-time	When the profile was created

Tasks

Field	Type / values	Notes
userId	uid	Owner (learner)
title	string	Task name
category	string	e.g. frontend, backend
dueDate	date	YYYY-MM-DD
priority	low medium high	Validated by security rules
status	todo in-progress done	Used to calculate progress

Bookings

Field	Type / values	Notes
learnerId	uid	Who requested support
assessorId	uid or null	Assigned assessor
topic	string	Help topic
preferredDate	ISO date-time	Requested session time
status	pending confirmed completed cancelled	Updated by the assessor
notes	string	Optional extra detail

Source files: database/schema.json (sample data) and database/database.rules.json (access rules).